# Sam Cheng-Tse Huang

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## EDUCATION

**Rutgers University – New Brunswick (RUNB)** — New Jersey, US
*Ph.D. Candidate in Physics* — *Sep. 2023 – Present*

**National Taiwan University (NTU)** — Taipei, Taiwan
*M.Sc in Physics* — *Sep. 2020 – July 2022*

**National Tsing Hua University (NTHU)** — Hsinchu, Taiwan
*B.Sc in Physics* — *Sep. 2016 – June 2020*

## PUBLICATIONS

[1] **A Universal Distribution of Dark Matter in Milky Way-like galaxies and How to Infer It**
**Huang, Sam Cheng-Tse**, Matthew R. Buckley, Justin I. Read, David Shih
(July 2026)

[2] **Updated constraints on the Georgi-Machacek model and its electroweak phase transition and associated gravitational waves**
Ting-Kuo Chen, Cheng-Wei Chiang, **Cheng-Tse Huang**, Bo-Qiang Lu
*Phys. Rev. D* 106 (5 Sept. 2022)p. 055019American Physical Society
DOI: 10.1103/PhysRevD.106.055019

[3] **Measurements of gluon fusion and vector-boson-fusion production of the Higgs boson in $H \to WW^* \to e\nu\mu\nu$ decays using $\mathrm{pp}$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector**
ATLAS collaboration
*ATLAS-CONF-2021-014* (Mar. 2021)

[4] **Consistency of Gauged Two Higgs Doublet Model: Gauge Sector**
**Cheng-Tse Huang**, Raymundo Ramos, Van Que Tran, Yue-Lin Sming Tsai, Tzu-Chiang Yuan
*JHEP* 09 (Sept. 2019)p. 048
DOI: 10.1007/JHEP09(2019)048

## PRESENTATIONS

**2nd UNDARK workshop** — Tenerife, Spain
*Joint Spatial–Spectral Gaussian Process Modeling of the Galactic Center Excess* — *Sep. 2026*

**Dark Matter and Stars** — Southampton, UK
*A Universal Distribution of Dark Matter in Milky Way-like galaxies and How to Infer It* — *July 2026*

**2026 Phenomenology Symposium** — Pittsburgh, USA
*Joint Spatial–Spectral Gaussian Process Modeling of the Galactic Center Excess* — *May 2026*

**International Joint Workshop on the Standard Model and Beyond** — Taipei, Taiwan
*Inferring DM Scales from Stellar Kinematic Distributions in MW-like Galaxies* — *Dec. 2025*

**Annual Meeting of the Physical Society of Taiwan** — Kaohsiung, Taiwan
*Updated constraints on the GM model and its EWPT and associated GWs* — *Jan. 2023*

**Asia-Pacific Workshop on Particle Physics and Cosmology** — Online
*EWPT in the Georgi-Machacek model under constraints of LHC data* — *Aug. 2021*

## WORK EXPERIENCES

**New High Energy Theory Center (NHETC) at RUNB** — New Jersey, US
*- Advisor: Dr. Matthew R. Buckley* — *Sep. 2023 – present*

**High Energy Phenomenology Group at NTU** — Taipei, Taiwan
*- Advisors: Dr. Cheng-Wei Chiang* — *Aug. 2022 – March 2023*
*- Advisor: Dr. Cheng-Wei Chiang* — *July 2020 – July 2022*

**ATLAS HWW Group at NTHU** — Hsinchu, Taiwan
*- Advisor: Dr. Pai-Hsien Jennifer Hsu* — *June 2019 – June 2020*

**Theoretical High Energy Group at Academia Sinica** — Taipei, Taiwan
*- Advisors: Dr. Yue-Lin Sming Tsai & Dr. Tzu-Chiang Yuan* — *June 2018 – June 2019*